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Advancing gene feature selection: a synergistic approach with co-expression networks and genetic algorithms

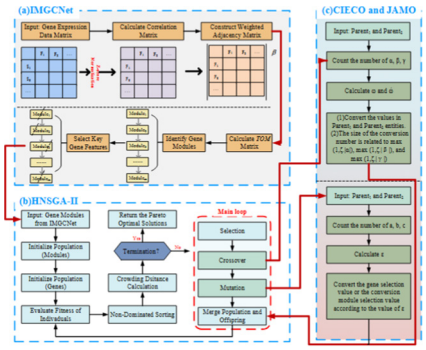
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HIGHLIGHTS

- CJWGA is proposed for gene feature selection to tackle the issues of high-dimensional medical gene data.
- CJWGA integrates co-expression networks and genetic algorithms, with IMGCNet for preprocessing module gene screening.
- CIECO and JAMO are designed in the genetic algorithm of CJWGA to balance intensification and diversification.
- Experimental results show that CJWGA can significantly reduce feature subset size while achieving high predictive accuracy across multiple gene datasets.

GRAPHICAL ABSTRACT



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ABSTRACT

Introduction: Gene feature selection is essential in bioinformatics and medical research, as it identifies gene subsets closely associated with specific diseases or biological traits from high-dimensional gene datasets. High-dimensional gene data causes the curse of dimensionality, leading to sparsity, complex inter-feature relationships, and significant noise. These issues undermine the reliability of conventional statistical methods in capturing underlying biological information. While gene feature selection can enhance classification model accuracy and reduce computational complexity, existing methods often struggle to handle the complexities of high-dimensional medical gene data.
Objectives: To address the limitations of current methods, we designed a synergistic gene feature selection approach (CJWGA) that integrates co-expression networks and genetic algorithms. The goal is to efficiently perform gene feature selection, significantly reduce the size of feature subsets, and achieve high predictive accuracy across multiple gene datasets.
Methods: The CJWGA decomposes feature selection into two key steps: preprocessing of co-expression networks and iterative selection using genetic algorithms. A preprocessing gene selection approach

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(IMGCNet) is proposed to screen module genes based on conditional mutual information. For joint mutual information, a combined information entropy crossover operator (CIECO) and a joint adaptive mutation operator (JAMO) are designed for the nondominated sorting genetic algorithm, aiming to balance intensification and diversification.

Results: Experimental results demonstrate that the proposed CJWGA achieves remarkable performance. It significantly reduces the size of feature subsets while maintaining high predictive accuracy across multiple gene datasets.

Conclusion: Overall, the synergistic CJWGA approach, integrating co-expression networks and improved genetic algorithms, exhibits excellent performance in gene feature selection. It addresses the challenges of high-dimensional medical gene data and holds potential as a valuable tool for gene feature selection in bioinformatics and medical research.

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Introduction

In modern medical research, the rapid advancements in biotechnology, particularly in high-throughput sequencing [1] and mass spectrometry [2], have led to an explosive increase in the volume of data generated in medical studies. High-dimensional data encompasses a variety of biological markers, including genomics [3], proteomics [4], and metabolomics [5]. Analyzing these data provides multi-layered insights into the onset and progression of diseases [6]. The integration and analysis of this data not only elucidate the molecular mechanisms underlying diseases but also identify potential biomarkers, serving as a basis for personalized treatment. Furthermore, the integrative analysis of high-dimensional data fosters the combination of multi-omics data, thus advancing the field of systems biology [7,8]. Systems biology emphasizes a holistic understanding of biological systems. By integrating data from different levels, such as genomics [9], transcriptomics [10], proteomics [11], and metabolomics, researchers can achieve a comprehensive understanding of disease development [12]. This integration aids in identifying critical biological pathways and molecular mechanisms, while also providing crucial clues for the development of novel therapies. For instance, in the study of metabolic diseases, integrating data from various omics can reveal metabolic pathway abnormalities, offering new targets for therapeutic strategies. Therefore, the effective interpretation of complex high-dimensional medical data enables researchers to gain deeper insights into the molecular mechanisms of diseases, thus propelling the development of personalized and precision medicine.

The effective analysis of high-dimensional data presents an urgent challenge, as the abundance of features can lead to the "curse of dimensionality" [13–15]. The intricate interrelationships among dimensions introduce uncertainties into the analysis of pathological data [16,17]. Typically, the number of genes in a dataset far exceeds the number of samples, resulting in sparsity [18,19] and complicating analysis and modeling. Additionally, the complexity of these data arises from the interplay of various biological factors, including gene-gene interactions, environmental influences, and individual variability. This complex structure often renders conventional statistical analyses inadequate for capturing underlying biological information. Moreover, high-dimensional gene data usually come with significant noise and uncertainty, which can stem from limitations in experimental techniques, sample processing, or individual differences, thereby challenging the reliability of the data. There may also be redundancy and correlation among the data, as many genes are functionally related, potentially leading to the emergence of redundant features that affect the accuracy of analysis results [20,21]. For example, in cancer research, researchers often collect extensive gene expression data to analyze the characteristics of tumors in different patients. In a hypothetical high-dimensional gene expression dataset, the expression levels of 10,000 genes may be measured. However, not all genes are directly related to tumor development or progression. Some genes may exhibit expression changes unrelated to tumors, or their expression may be influenced by other factors (such as environmental or individual differences) rather than tumor characteristics. Additionally, the relationships between genes and phenotypes are often nonlinear, indicating that traditional linear models may not effectively capture these complex relationships, necessitating the use of more sophisticated models for analysis. Lastly, modern high-throughput sequencing technologies enable the generation of large-scale gene expression data, further increasing data dimensionality and complexity. Fig. 1 illustrates the relationship between the number of gene features, accuracy, and runtime. As the number of features increases, the computational complexity of the model generally escalates, resulting in significantly longer runtimes. Moreover, an increase in feature quantity does not necessarily correlate with improved model accuracy. Conversely, excessive features may introduce redundancy and noise [22,23], leading to model overfitting, which performs well on training data but exhibits diminished generalization capability on test data.

Therefore, feature selection is a critical step in ensuring the validity and reliability of research in high-dimensional medical gene data analysis [24,25]. By implementing feature selection techniques, researchers can identify and retain key genes that significantly impact disease mechanisms and prognosis, thereby eliminating redundant information and noise. This process not only helps reduce data dimensionality, alleviating computational burdens and enhancing model efficiency, but also improves predictive accuracy and strengthens generalization ability [26–28]. Furthermore, feature selection can provide clear biological interpretations, revealing potential mechanisms of disease and offering important biomarkers for clinical applications. Consequently, feature selection is not only a necessary step in data processing but also a vital foundation for achieving precision medicine and personalized treatment.

However, a medical gene dataset with n features can generate 2^n feature subsets, making it impractical to explore all subsets in high-dimensional space due to this exponential growth [29]. Common feature selection methods include filtering [30], wrapper [31], and embedded approaches [32]. Filtering methods rely on specific evaluation metrics for feature selection, such as mutual information theory [33], which are efficient in high-dimensional data but

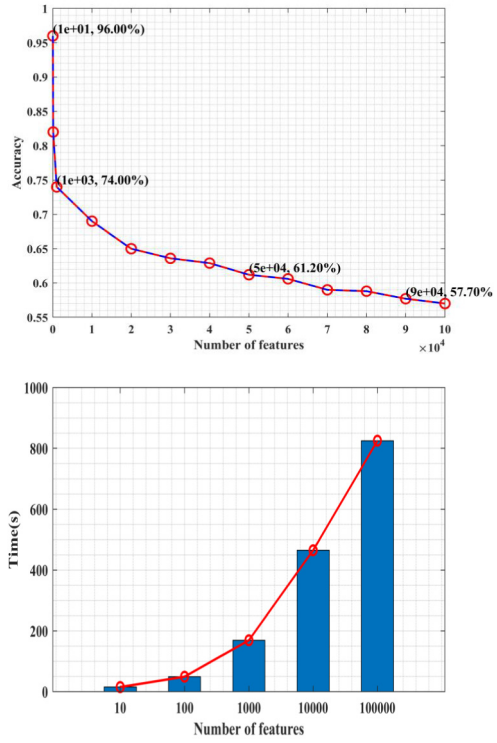


Fig. 1. The balance relationship between the number of features, model accuracy and running time. The above figure illustrates that too many features will introduce redundancy and noise, resulting in overfitting of the model rather than improving accuracy. The figure below shows that as the number of features increases, the computational complexity rises and the running time prolongs.

fail to capture interactions and complex joint relationships among features. Wrapper methods evaluate feature subsets by passing them to specific learning algorithms, selecting optimal features based on model performance, typically relying on cross-validation to assess model accuracy. This approach can identify complex nonlinear relationships and interactions among features with high accuracy, albeit at a greater computational cost. Embedded methods combine the advantages of filtering and wrapper methods by automatically selecting features during model training. These methods are often associated with specific algorithms, such as LASSO regression [34] and tree-based models [35], which adjust feature weights during training to achieve automatic selection.

While filtering, wrapper, and embedded methods each have their strengths, they also exhibit notable limitations when addressing complex high-dimensional gene data. These methods primarily focus on dimensionality reduction without considering that the analysis of medical data also involves studying gene pathways, signaling networks, and biomarkers—biological factors crucial for understanding disease mechanisms and developing treatment strategies. To address this issue, Zhang et al.[36] proposed Weighted Gene Co-expression Network Analysis (WGCNA) based on the theory of biological genetics. WGCNA constructs a co-expression network among genes, enabling the identification of cooperative expression relationships [37,38]. In biological systems, genes do not operate in isolation; rather, they participate in biological processes through complex interactions. WGCNA allows researchers to understand these interactions from a systems perspective, identifying critical regulatory networks under specific biological conditions. Additionally, WGCNA clusters genes into functionally related modules, aiding researchers in comprehensively understanding biological processes. Moreover, WGCNA can help identify gene modules associated with specific phenotypes or disease states, providing clues for potential biomarkers. The biological-driven nature of this method enables researchers to pinpoint key genes related to diseases within large-scale genomic data [39,40].

Despite the significant advantages of WGCNA in analyzing high-dimensional medical data, it does have limitations that may affect its application in specific contexts. Firstly, WGCNA primarily constructs co-expression networks based on linear correlations, which may not adequately capture nonlinear relationships among genes. This limitation could result in the omission of important gene interactions in complex biological processes. Secondly, WGCNA assumes that modules are relatively independent during module partitioning; however, interactions and dependencies between modules in real biological systems can be quite intricate, potentially impacting the comprehensive understanding of biological processes. The shortcomings of wrapper methods arise from their high computational costs. WGCNA can effectively extract a small number of core modules and key genes, and based on the aforementioned methods, this paper proposes a Weighted Non-dominated Sorting Genetic Algorithm (CJWGA) based on Combined Information Entropy Crossover Operator and Joint Adaptive Mutation Operator to address the feature selection problem in complex high-dimensional gene medical data. The objectives that require optimization are two fold: the number of selected features and the accuracy of the model. Our aim is to achieve excellent classification performance while selecting the smallest subset of features in gene data.

The main contributions of this paper are as follows:

- A synergistic approach with co-expression networks and genetic algorithms CJWGA is proposed for feature selection in complex high-dimensional gene medical data. This method can effectively select key gene features to achieve accurate classification tasks.
- Based on conditional mutual information theory, an IMGCNet method is proposed to realize the initial effective selection of high-dimensional gene data, which can effectively reduce the size of the initially generated feature set.
- The Combined Information Entropy Crossover Operator (CIECO) and Joint Adaptive Mutation Operator (JAMO) are introduced to achieve a better balance between intensification and diversification in gene feature selection tasks.
- The CJWGA algorithm demonstrates excellent performance in high-dimensional gene feature selection.

The rest of this paper is structured as follows: Section 2 describes the related work of other researchers. In Section 3, the background knowledge of feature selection in genetic data is introduced. Section 4 describes in detail the feature selection method proposed in this paper. In Section 5, the overall performance of the CJWGA method is verified by various experiments. Finally, Section 6 summarizes the work and looks forward to the future research direction.

Related work

The continuous emergence of high-dimensional data has led to an increasing preference for filtering methods, which are favored

for their simplicity and ease of design, as well as their lower complexity [41,42]. Solorio-Fernández et al. [43] developed a supervised filtering feature selection method that combines information-theoretic redundancy analysis to identify and process relevant and redundant features within a dataset. Andrea Bommert et al. [44] experimentally demonstrated that simple variance filtering methods can exhibit superior performance in high-dimensional feature selection. Obaido et al. [45] proposed a filter-based feature selection framework that leverages the collective advantages of multiple base models' predictions. Xue et al. [46] developed two multi-objective filter-based feature selection frameworks that utilize mutual information and entropy as distinct evaluation criteria, resulting in four multi-objective feature selection methods that were tested for classification performance using decision trees.

However, in pursuit of improved classification accuracy, many researchers have adopted traditional wrapper methods for feature selection. A typical process involves subset generation, subset evaluation, and stopping criteria. Candidate subsets are evaluated based on certain standards and compared with the previous best subset; if the newly generated subset outperforms the previous one, it becomes the latest best subset. This process continues until a predetermined stopping criterion is reached. Medjahed et al. [47] employed a Binary Dragonfly Algorithm (BDA) for feature selection, achieving commendable classification accuracy across six microarray datasets. Luo et al. [48] introduced a novel approach to eliminate irrelevant and redundant features in feature selection, combining the rough hyperrectangle algorithm with a binary particle swarm optimization algorithm integrated into a cloud computing model. Experimental results indicated that this method outperformed several outstanding feature selection optimizers in terms of classification accuracy, the number of selected features, and computation time. Hashemi et al. [49] proposed a multi-criteria decision-making ant colony algorithm that draws on the judgments of various experts, utilizing different heuristic methods to gather more information at subsequent nodes. These strategies effectively enhanced the performance of the ant colony algorithm, yielding more competitive results compared to several well-known feature selection algorithms. Ibrahim et al. [50] combined the Slap algorithm with particle swarm optimization (PSO) to improve the exploration and exploitation capabilities of PSO, achieving high accuracy on the UCI dataset. Inbarani et al. [51] noted the typical small sample sizes and high feature dimensions in medical data, proposing a new method that integrates relative reduction based on particle swarm optimization with fast reduction hybrid approaches, validating its effectiveness across multiple standard datasets. Chen et al. [52] hybridized genetic algorithms and K-medoids methods, using distance similarity to group genes, and achieved excellent results on public brain cancer datasets with this hybrid method. Derrac et al. [53] proposed a cooperative co-evolution model combined with nearest neighbor methods for data reduction. Mukhopadhyay and Maulik [54] integrated multi-objective genetic algorithms with support vector machine classifiers to evaluate the chromosomes of encoded subsets.

Although these wrapper methods generally yield better results than filtering methods, they are slower and incur higher computational costs due to their reliance on the resource demands of modeling algorithms [55]. Consequently, combining both approaches is also a viable option. Nematzadeh et al. [56] designed a two-stage hybrid feature selection method—maximal pattern recognition—multi-objective discrete evolution strategy, which significantly outperformed existing frequency-based rankers in identifying nonlinear patterns while effectively eliminating most non-informative features. Xu et al. [57] proposed a hybrid feature selection algorithm combining filtering and population evolution (FG-HFS) for high-dimensional gene expression data, grouping redundant features using spectral clustering and filtering them based on the Markov covering principle, followed by multi-objective genetic algorithms for the subsequent selection of the filtered feature subset. Mandal et al. [58] aimed to obtain the optimal feature subset from high-dimensional, small-sample datasets by integrating five filtering methods with a differential evolution algorithm for effective dimensionality reduction. Bir-Jmel et al. [59] considered potential interactions among genes by searching for gene subsets that maximize relevance and minimize redundancy (MRMR), subsequently combining variable neighborhood search (VNS) metaheuristics with innovative stochastic local search (SLS) algorithms to identify the final gene subset.

Some researchers have sought effective feature extraction from a biological perspective. Wong and Burkowski [60] proposed a new feature selection algorithm for QSAR studies based on kernel alignment, which serves as a measure of similarity between two kernel functions and employs recursive feature elimination to calculate the molecular descriptors containing the most significant features for classification applications. Wang et al. [61] introduced a method for pre-filtering microarray biological data based on background knowledge, utilizing L1 logistic regression techniques for feature extraction. The results indicated that the pre-filtering approach could significantly reduce the number of features in mRNA and microRNA expression datasets.

In summary, filter-based methods utilize statistical or correlation analysis to rapidly select features. However, they often fail to account for interactions between features, which may result in the omission of critical ones. Wrapper-based methods, on the other hand, integrate specific learning algorithms to directly optimize model performance. While achieving higher accuracy, these methods are computationally expensive and prone to overfitting. Some studies have attempted to combine the two approaches or incorporate medical domain knowledge for feature identification. Nonetheless, these strategies often lack systematic frameworks or fail to fully explore the complex structural information inherent in the data.

In contrast to previous work, we employed a WGCNA for the preliminary screening of medical gene data and utilized a multi-objective evolutionary algorithm to select the main modules identified. This approach considers the interactions among modules, while the selection of specific genes within modules relies on two information-theoretic crossover and mutation operators, enabling effective selection of the most informative and representative key genes across the entire dataset.

As summarized in Table 1, existing feature selection methods either focus on statistical relevance or rely on search strategies guided by classification performance. However, most studies do not fully exploit the biological structural relationships among genes. In contrast, the proposed CJWGA integrates co-expression network analysis with multi-objective evolutionary optimization, enabling the selection of biologically meaningful gene subsets while maintaining strong classification performance.

Table 1
Comparison of representative gene feature selection methods.

Type	Main Idea	Advantages	Limitations
Filter	Select features based on information theory	Fast and scalable	Ignores feature interactions
Wrapper	Uses swarm intelligence to search optimal subset	High classification accuracy	Computationally expensive
Hybrid	Combine filtering and evolutionary search	Balance efficiency and accuracy	Ignore biological structure
CJWGA	WGCNA preprocessing + genetic algorithm	Considers gene co-expression relationships and feature interactions	Higher algorithmic complexity

Background knowledge

In gene medical feature selection, optimization problems typically involve two primary goals: minimizing the number of features and maximizing model accuracy. We can formalize the feature selection problem as a multi-objective optimization model, aiming to identify Pareto optimal solutions that balance feature selection with model performance. Let the feature set be denoted as $F = \{x_1, x_2, \ldots, x_n\}$ and define the decision variable for feature selection as $x_i (i = 1, 2, \ldots, n)$, where $x_i = 1$ indicates that feature is selected, while $x_i = 0$ indicates that it is not selected. The following two objective functions can be defined:

1. The size of the selected feature subset:

$$f_1(X) = |S| = \sum_{i=1}^{n} x_i \tag{1}$$

2. Model accuracy:

$$f_2(X) = Accuracy = \frac{\sum_{i=1}^{N} 1(Y_i = T_i)}{N} \tag{2}$$

where, S is the number of features in the selected feature subset, Y and T are the classification results of the real label and model, respectively, and N is the number of samples. Our goal is to find the feature selection vector X such that:

$$\begin{cases} Minimize & f_1(X) \\ Maximize & f_2(X) \\ Subject\ to & x_i \in \{0, 1\}, \forall i \in \{1, 2, \ldots, n\} \end{cases} \tag{3}$$

In this paper, to present the results more intuitively, the accuracy is expressed in terms of error rate.

Proposed algorithm

Aim at complex high-dimensional gene data, a synergistic approach CJWGA is designed for gene feature selection, which integrates co-expression networks and genetic algorithms. 1) Based on WGCNA and conditional mutual information theory, IMGCNet is proposed to partition module genes and extract the most crucial gene features. 2) The extracted modules and gene features are hierarchically encoded using the nondominated sorting genetic algorithm. 3) An improved version of the nondominated sorting genetic algorithm, incorporating a combined information entropy crossover operator CIECO and joint adaptive mutation operator JAMO is proposed for a better balance between intensification and diversification in gene feature selection.

IMGCNet

Algorithm 1: IMGCNet

Input: *Data*: Gene expression data matrix;
F: Number of gene features;
Output: Gene module M and main characteristics of the module

/* *Main loop*/
for $j = 1$ **to** F **do**
  $Z_j \leftarrow (x_j - \overline{x}_j)/s_j$;
end
/* *Perform* $z-score$ *transformation on the data;* */
Calculate the correlation matrix R between genes using Eq. 5;
Select soft threshold parameters β;
for $i = 1$ **to** F **do**
  for $j = 1$ **to** F **do**
    $A_{ij} \leftarrow \left|r_{ij}\right|^{\beta}$;
  end
end
for $i = 1$ **to** F **do**
  for $j = 1$ **to** F **do**
    $TOM_{ij} \leftarrow (a_{ij} + \sum_k a_{ik} \cdot a_{jk})/min(\sum_k a_{ik}, \sum_k a_{jk})$;
  end
end
for i *in* $1 : M$ **do**
  Calculate the module eigenvector x_k of M_i;
  Calculate the number of genes N_i for M_i;
  for j *in* $1 : N_i$ **do**
    Calculate the $I(x_j; y|x_k)$;
  end
  The top $(M + N_i - 1)$ genes in the $I(x_j; y|x_k)$ index were preserved according to the size of N_i;
end
return M, x_k and key gene M_{x_i}.

In genomic research, elucidating the relationships between genes and their associations with biological phenotypes is crucial for understanding complex biological processes. Due to the high dimensionality and noisy nature of gene expression data, traditional feature selection methods often struggle to accurately

identify truly relevant genes. However, the interactions between genes can be represented using networks, making it feasible to group genes using WGCNA. WGCNA effectively handles complex gene expression data by constructing a weighted co-expression network that reveals the relationships among genes and identifies biologically relevant gene modules. In simpler terms, a module consists of a set of genes that are highly correlated within the co-expression network and may play significant roles in the same biological processes. This approach not only considers direct interactions between genes but also integrates shared connections through metrics such as Topological Overlap Measure (TOM), enhancing the reliability and stability of the network. Furthermore, WGCNA can guide potential biomarker discovery and functional studies by correlating gene modules with external phenotypes. Assuming we are partitioning gene modules for a dataset, Data, containing N samples and n gene features, Z-score normalization is necessary to eliminate the effects of varying gene expression levels. Specifically, the standardized expression value for the jth gene can be expressed as:

$$Z_j = \frac{x_j - \overline{x}_j}{s_j} \tag{4}$$

where, x_j represents the raw expression value of gene j in the samples, while $\overline{x}_j$ and s_j denote the mean and standard deviation of that gene, respectively. Next, a similarity matrix R is constructed by calculating the Pearson correlation coefficients between genes, where each element r_{ij} indicates the correlation between gene i and gene j.

$$r_{ij} = \frac{\sum_{k=1}^{N}(Z_{ik} - \overline{Z}_i)(Z_{jk} - \overline{Z}_j)}{\sqrt{\sum_{k=1}^{N}(Z_{ik} - \overline{Z}_i)^2}\sqrt{\sum_{k=1}^{N}(Z_{jk} - \overline{Z}_j)^2}} \tag{5}$$

Next, the similarity between genes is defined using the weighted adjacency matrix A.

$$A_{ij} = |r_{ij}|^{\beta} \tag{6}$$

where, β is a soft thresholding parameter that is typically determined by analyzing the scale-free of the network. In this paper, the pickSoftThreshold function is used to test the candidate β values, and the minimum value that first achieves an R^2 of 0.90 for the scale-free topological fitting is selected as the optimal soft threshold. A scale-free network is characterized by a degree distribution that follows a power law, exhibiting a few highly connected nodes while the majority of nodes have fewer connections. This phenomenon arises because, within biological regulatory networks, only a small number of genes play crucial regulatory roles, while other genes have comparatively lower regulatory hierarchies. The advantage of this type of network is that the overall functionality of the organism remains intact even if some less critical peripheral genes are disrupted. After processing, the few strongly correlated gene features are either unaffected or minimally impacted, while the correlations of weakly correlated features significantly decrease when raised to the power of β. This design aligns more closely with biological significance. The key parameters involved include the module size and the height of the tree diagram cut-off. The former is set at 30, while the latter is 0.25, corresponding to a correlation threshold of approximately 0.75 among the characteristic genes of the modules. These parameter settings ensure that the identified modules have biological significance and statistical reliability. Fig. 2 presents an example of a scale-free network. After constructing the adjacency matrix, the Topological Overlap Measure *TOM* is calculated to assess the similarity between genes. The formula for computing the TOM matrix is as follows:

$$TOM_{ij} = \frac{A_{ij} + \sum_k A_{ik} \cdot A_{jk}}{min\left(\sum_k A_{ik}, \sum_k A_{jk}\right)} \tag{7}$$

The *TOM* reflects the similarity between gene i and gene j while taking into account their common neighbors. Subsequently, the *TOM* matrix is converted into a dissimilarity matrix, and hierarchical clustering algorithms are applied to the *TOM* matrix to identify k gene modules and determine their module eigengene. The module eigengene are defined as the result of principal component analysis conducted on all genes within the module, with the first principal component representing the module eigengene. This value reflects the overall expression level (expression pattern) of the genes within the module; thus, the module can be regarded as a gene, and the module eigengene can be considered the expression value of this gene. However, relying solely on the first principal component may result in the loss of specific information and functions of other genes within the module. A single eigenvalue gene cannot comprehensively reflect the biological functions of the module or its complex relationship with phenotypes. Another potential issue is bias, where the selected module eigengene may be influenced by particular gene expression patterns, leading to an underestimation of the importance of other genes within the module. For example, if the expression level of a particular gene is abnormally high or low, it may dominate the calculation of the eigenvalue gene, despite not being the functionally most significant gene. Additionally, due to biological diversity, the genes within a module may exhibit varied functions and complex interactions. Selecting a single eigenvalue gene fails to represent this diversity, potentially leading to misunderstandings of biological processes. It is also essential to consider the relationships not only between modules but also between gene features and labels. Therefore, a more effective evaluation method is required to select more representative gene features. In this context, conditional mutual information was chosen to further select gene features. For a module $M_i = \{x_1, x_2, x_3, \ldots x_m\}$, let x_k be the eigenvalue gene obtained in the previous step. The goal is to determine the information dependence of x_i (where $i = 1, 2, 3, \ldots, k-1, k+1, \ldots, m$) on the Label Y, given x_k. A higher value of conditional mutual information indicates that knowing the state of one variable provides more information about the state of another variable. The formula for calculating mutual information is as follows:

$$I(X;Y|Z) = \sum_{x\in X}\sum_{y\in Y}\sum_{z\in Z} P(x,y,z) log \frac{P(x|y,z)}{P(x|z)} \tag{8}$$

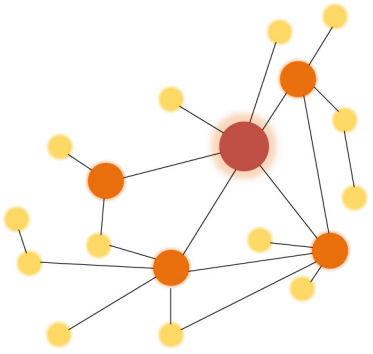


Fig. 2. Scale-free networks are complex networks where the degree distribution of nodes follows a power-law pattern. In these networks, a small number of nodes have a large number of connections, while the majority of nodes have only a few connections.

where, $P(x,y,z)$ represents the joint probability distribution of X, Y and Z, while $P(x|y,z)$ denotes the conditional probability distribution of X given Y and Z. Additionally, $P(x|z)$ represents the conditional probability distribution of X given Z. Therefore, a higher value of conditional mutual information indicates that, even when controlling for the conditional variable Z, there remains a strong dependency between X and Y.

This metric considers the combinatorial relationships between gene features within modules and their corresponding labels, allowing for the calculation of conditional mutual information values for all non-feature genes. When selecting genes between modules, it is important to consider the size of the modules, as the number of genes typically reflects the diversity and complexity of the biological processes within. Modules with a larger number of genes often encompass a variety of functionally related genes, which may collaboratively play roles in different biological processes. To effectively capture this diversity and complexity, selecting a greater number of genes can provide a more comprehensive representation of the module's functions, aiding in the identification of correlations with specific biological processes or phenotypes. Conversely, modules with fewer genes may focus on specific biological functions or regulatory networks. In such cases, retaining too many genes could introduce unnecessary noise or interference, thereby impacting the understanding of the core functions of the module. Thus, appropriately reducing the number of genes can enhance the focus of the analysis and improve the biological significance of the results. Assuming there are m modules $(M_1, M_2, \ldots, M_m)$, with each module M_i containing N_i genes, the modules are sorted in descending order based on gene count, satisfying $N_1 > N_2 > \ldots > N_m$. The calculation method for the number of retained genes, denoted as C_i, is defined as $C_i = max(0, M + 1 - i)$. This approach ensures that larger modules retain more genes, while smaller modules correspondingly retain fewer genes, thereby helping to maintain the biological representativeness and informational content between modules. Algorithm 1 represents the running flow of the IMGCNet model.

HNSGA-II

After performing WGCNA, we typically obtain multiple gene modules that represent the co-expression relationships among genes under specific biological processes or conditions. However, the number of modules generated by WGCNA can be substantial, and preliminary analyses often do not adequately consider the interactions between these modules. This oversight may hinder a clear understanding of the mutual influences and overall effects among the modules in subsequent biological research. Employing a combinatorial approach to select modules typically requires evaluating all possible combinations, leading to an exponential increase in the number of combinations as the number of modules grows. Consequently, this results in a dramatic increase in computation time, particularly with large-scale gene expression datasets, which can consume considerable time and computational resources. Similarly, using traditional selection methods or simple heuristic algorithms for module selection may focus solely on the currently optimal modules, overlooking the globally best combination, thus yielding suboptimal results. Selecting all modules may lead to the retention of redundant information, complicating the analysis. Moreover, retaining too many modules may obscure important biological signals, reducing the interpretability of the results.

Algorithm 2: Crossover operator

Input: $ProC$: Crossover probability ;
η_c: Distribution index;
D: Dimension of a problem;
Pa_1, Pa_2: Parent individual;
Output: $Offspring_1, Offspring_2$

```
/* Binary crossing*/
if ProC ≤ rand then
    Pa_1 ← Pa_1;
    Pa_2 ← Pa_2;
else
    for d in 1 : D do
        u ← rand(0, 1);
        if u ≤ 0.5 then
            u* ← (2 · u)^{1/(η_c+1)};
        else
            u* ← (1/(2 · (1 − u)))^{1/(η_c+1)};
        end
        Pa_1(d) ←
          0.5 · ((1 + u*) · Pa_1(d) + (1 − u*) · Pa_2(d));
        Pa_2(d) ←
          0.5 · ((1 − u*) · Pa_1(d) + (1 + u*) · Pa_2(d));
    end
end
Offspring_1 ← Pa_1 ;
Offspring_2 ← Pa_2 ;
return Offspring_1, Offspring_2.
```

Algorithm 3: Mutation operator

Input: $ProM$: Mutation probability;
η_m: Distribution index;
D: Dimension of a problem;
Pa_1, Pa_2: Parent individual;
x_{max}: Upper bound of value;
x_{min}: Lower bound of value;
Output: $Offspring_1, Offspring_2$

```
/* Polynomial variation*/
if ProM ≤ rand then
    Pa_1 ← Pa_1;
    Pa_2 ← Pa_2;
else
    for d in 1 : D do
        v ← rand(0, 1);
        if v ≤ 0.5 then
            v* ← (2 · v)^{1/(η_m+1)} − 1;
        else
            v* ← 1 − (2 · (1 − v))^{1/(η_m+1)};
        end
        Pa_1(d) ← Pa_1d + v* · (x_max(d) − x_min(d));
        Pa_2(d) ← Pa_2d + v* · (x_max(d) − x_min(d));
    end
end
Offspring_1 ← Pa_1 ;
Offspring_2 ← Pa_2 ;
return Offspring_1, Offspring_2.
```

Algorithm 4: HNSGA-II

Input: N: the number of population;
$MaxFEs$: Maximum evaluation times;
F: The number of genetic traits;
$ProC$: Crossover probability ;
$ProM$: Mutation probability;
M: Number of optimization objectives;
η_c, η_m: Distribution index;
NM: Number of modules;

Output: *Pareto-optimal front*

$P \leftarrow$ Initialization (N);
for i *in* $1 : NM$ **do**
  $P_i \leftarrow$ Initialization $(|Module_i|)$;
end
$FEs \leftarrow 0$;
/* *Main loop*/
while *NotTermination* $(FEs \leq MaxFEs)$ **do**
  Create offspring population $Q_t \leftarrow \left\{p_1^{'}, p_2^{'}, ..., p_N^{'}\right\}$ using Algorithm. 2 and Algorithm. 3;
  for j *in* $1 : M$ **do**
    Evaluate the f_j of individuals in Q_t;
  end
  Combine populations: $P_t \leftarrow P_t \cup Q_t$;
  for j *in* $1 : N$ **do**
    $d_j = \sum_{m=1}^{M} \frac{f_m^{(j+1)} - f_m^{(j-1)}}{f_m^{\max} - f_m^{\min}}$;
  end
  Perform non-dominated sorting on P_t to obtain Pareto fronts;
  Select the best N individuals to form the next generation P_{t+1};
end
return Pareto-optimal front of the last population P.

To address this issue, we selected the NSGA-II [62] for further module selection. As a multi-objective optimization algorithm, NSGA-II effectively handles trade-offs among multiple objectives. This approach allows us to consider not only individual functionalities when selecting modules but also the relationships between them, thereby enhancing the biological interpretability and practical utility of the analysis results. The advantages of using NSGA-II lie in its global search capabilities and its multi-objective handling mechanism. By maintaining diversity among solutions and identifying Pareto optimal solutions, we can obtain various potential combinations of modules, each with distinct advantages across different objectives. This provides broader options and deeper insights for subsequent experimental validation and biological research. The main process of NSGA-II is as follows: Step 1): Generate initial population randomly $P = \{p_1, p_2, \ldots p_N\}$, where N is the population size. It is important to note that here we initialize the module selection process, and the genes within each module will also undergo random initialization.

Step 2): For each individual p_i, we calculate the fitness values $f_j(p_i)$ across multiple objective functions, where j represents the index of the objective function.

Step 3): Perform a non-dominated sorting of the population, defining the condition under which individual p_i dominates individual p_j as follows: $p_i \prec p_j \Longleftrightarrow (\forall j, f_j(p_i) \leqslant f_j(p_j)) \wedge \exists k, f_k(p_i) < f_k(p_j)$. Individuals are classified into different ranks $F_1, F_2, F_3, \ldots, F_K$, where F_i represents the non-dominated solutions at the i_{th} level.

Step 4): For each individual p_j in the rank F_i, the crowding distance d_j is calculated. $d_j = \sum_{m=1}^{M} \left(f_j^{max}(m) - f_j^{mim}(m)\right)$. $f_j^{max}(m)$ and $f_j^{min}(m)$ represent the maximum and minimum fitness values of individual p_j on objective m, respectively. Step 5): Selection, crossover, and mutation operations: Use tournament selection based on non-domination rank and crowding distance to select the new population $Q = \{p_1', p_2', \ldots, p_N'\}$. Algorithm 2 and Algorithm 3 represent the crossover and mutation processes of NSGAII algorithm respectively. Step 6): Merge the parent population and the offspring population $P_t = P_t \cup Q_t$. Perform non-dominant sorting again and select the top N individuals from the population to form a new population. Continue iterating until the completion of the iterations, and output the final $ParetoFront = \{p \in P_{final} | \neq xistsy \in P_{final} s.t. y \prec p\}$.

Through IMGCNet, all gene data is partitioned into distinct, highly correlated internal modules, and key representative genes are extracted based on the conditional mutual information criterion. If these modules were encoded using a conventional NSGA-II approach, the previous operations would become meaningless, and the concept of modules would be rendered obsolete. Therefore, we propose a hierarchical encoding method: the first layer encodes the modules, while the second layer encodes genes within each module. Mutation and crossover in the iterative process operate alternately and independently (details are provided in Section 4.3). The complete flow of the HNSGA-II algorithm is shown in Algorithm 4.

Algorithm 5: Combined Information Entropy Crossover Operator (CIECO)

Input: N: the number of population;
M: Gene network module;
Output: $Offspring_1, Offspring_2$

/* *Combination information entropy crossover operator*;*/
$\alpha, \beta, \gamma \leftarrow 0$;
Select Pa_1 and Pa_2;
for *i in* $1 : \left|M_{Pa_1}\right|$ **do**
  if $(Pa_1(i) + Pa_2(i) == 2)$ **then**
    $\alpha \leftarrow \alpha + 1$;
  else
    if $(Pa_1(i) + Pa_2(i) == 0)$ **then**
      $\beta \leftarrow \beta + 1$;
    else
      $\gamma \leftarrow \gamma + 1$;
    end
  end
end
$I(x_{\alpha_1}, x_{\alpha_2}, ...) \leftarrow \sum H(\alpha_i) - H(\alpha_1, \alpha_2, ...)$;
$I(x_{\beta_1}, x_{\beta_2}, ...) \leftarrow \sum H(\beta_i) - H(\beta_1, \beta_2, ...)$;
$I(x_{\gamma_1}, x_{\gamma_2}, ...) \leftarrow \sum H(\gamma_i) - H(\gamma_1, \gamma_2, ...)$;
$\omega \leftarrow \frac{I(x_{\alpha_1}, x_{\alpha_2}, ...) \cdot |\beta| - I(x_{\beta_1}, x_{\beta_2}, ...) \cdot |\alpha|}{I(x_{\gamma_1}, x_{\gamma_2}, ...) \cdot |\gamma|}$;
$\omega' \leftarrow 1 - e^{-|\omega|}$;
if $\omega > 0$ && $\omega' > rand$ **then**
  Select $max(1, \zeta \cdot |\alpha|)$ x_{α_i} individuals and set their value to 0;
  Select $max(1, \zeta \cdot |\gamma|)$ x_{γ_i} individuals and set their value to 1;
else
  Select $max(1, \zeta \cdot |\alpha|)$ x_{α_i} individuals and set their value to 0;
  Select $max(1, \zeta \cdot |\beta|)$ x_{β_i} individuals and set their value to 1;
end
$Offspring_1 \leftarrow Pa_1$;
$Offspring_2 \leftarrow Pa_2$;
return $Offspring_1, Offspring_2$.

Algorithm 6: Joint Adaptive Mutation Operator (JAMO)

Input: N: the number of population;
F: The number of genetic traits;
M: Gene network module;
δ: Control parameter;
FEs: Current evaluation times;
$MaxFEs$: Maximum evaluation times;
Output: $Offspring$

/* Joint adaptive mutation operator;*/
Select Pa;
$b_1 \leftarrow 0$;
Calculate a, b, and c;
for *i in* $1 : \left|M_{Pa}\right|$ **do**
  if $M_{Pa}(i) == 1$ **then**
    $b_1 \leftarrow b_1 + 1$
  end
end
$a \leftarrow FEs/MaxFEs$;
$b \leftarrow b_1 / \left|M_{Pa}\right|$;
$\epsilon = e^{-(a^\delta - b^\delta + log(1+|c|))}$;
if $\epsilon < rand$ **then**
  $Pa = 1 - Pa$;
else
  Randomly selected module M_{Pa_1} within gene x_i transformation;
end
$Offspring_1 \leftarrow Pa$;
return $Offspring$.

Algorithm 7: CJWGA

Input: N: the number of population;
NSGA-II parameters;
δ: Control parameter;
Mo: Number of optimization objectives;
$MaxFEs$: Maximum evaluation times;
M: Gene network module;

Output: *Pareto-optimal front*

$P \leftarrow$ Initialization (N);
$FEs \leftarrow 0$;
/* *Main loop*/
while *NotTermination* ($FEs \leq MaxFEs$) **do**
  Create offspring population $Q_t \leftarrow \left\{p_1', p_2', ..., p_N'\right\}$ using Algorithm. 5 and Algorithm. 6;
  for j *in* $1 : Mo$ **do**
    Evaluate the f_j of individuals in Q_t;
  end
  Combine populations: $P_t \leftarrow P_t \cup Q_t$;
  for j *in* $1 : N$ **do**
    $d_i = \sum_{m=1}^{M}(f_j^{max}(m) - f_j^{mim}(m))$;
  end
  Perform non-dominated sorting on P_t to obtain Pareto fronts;
  Select the best N individuals to form the next generation P_{t+1};
end
return Pareto-optimal front of the last population P.

Through the aforementioned NSGA-II, we can identify several important modules. However, relying solely on its own crossover and mutation operations for feature selection is clearly inadequate for genomic data. On one hand, NSGA-II heavily depends on crossover and mutation for generating offspring; since we are encoding entire modules, the retention and transmission of high-quality genes within those modules may be insufficient, which can adversely affect the quality of feature selection. On the other hand, the importance of genes within different modules can vary significantly, and NSGA-II's crossover and mutation operations may not adequately account for this, leading to the oversight or misselection of crucial gene features. Therefore, we propose the combination information entropy crossover operator and joint adaptive mutation operator.

When applying NSGA-II to feature selection in genomic data, the crossover operation simply involves exchanging two values at a given position. In this context, we encode the modules rather than the specific genes within those modules. Consequently, even if module M_i is selected by two individuals p_i and p_j, the specific gene selections within that module are likely to differ significantly. Therefore, designing an effective method to evaluate the internal genes can substantially enhance the model's performance. Here, we introduce the concept of joint mutual information, which measures the degree of information sharing between two or more random variables. Specifically, it quantifies how much knowing one variable reduces the uncertainty about another variable. The calculation is as follows:

$$I(X_1;X_2;\ldots;X_n) = H(X_1) + \ldots + H(X_n) - H(X_1,\ldots,X_n) \tag{9}$$

in which, $I(X_1;X_2;\ldots;X_n)$ represent the joint mutual information between multiple random variables $X_1, X_2, \ldots X_n$, while $H(X_i)$ denotes the entropy of a single variable X_i, reflecting the uncertainty associated with that variable. The term $H(X_1, X_2, \ldots, X_n)$ signifies the joint entropy of all variables, indicating their collective uncertainty. The calculation of $H(X)$ is expressed as $H(X) = -\sum_{i=1}^{n} P(x_i) log P(x_i)$, where n is the number of possible values that the random variable X can take, x_i represents each possible value of X, and $P(x_i)$ denotes the probability of x_i occurring.

The calculation for $H(X_1, \ldots, X_n)$ is represented as $-\sum_{i,j,\ldots,k} P(x_i, x_j, \ldots, x_k)\ log P(x_i, x_j, \ldots, x_k)$, where $P(x_i, x_j, \ldots, x_k)$ is the joint distribution probability of these variables.

From the previous discussion, it can be concluded that the features retained in the crossover step module are those that minimize label uncertainty to the greatest extent. However, further processing of the genes within the module is necessary to identify the smallest number of key genes that most effectively represent the characteristics of the module. Consequently, the joint mutual information entropy crossover operator has been proposed, expressed as follows:

$$\omega = \frac{I(x_{\alpha_1}, x_{\alpha_2}, \ldots) \cdot |\beta| - I(x_{\beta_1}, x_{\beta_2}, \ldots) \cdot |\alpha|}{I(x_{\gamma_1}, x_{\gamma_2}, \ldots) \cdot |\gamma|} \tag{10}$$

where, ω represent the value of the joint mutual information entropy crossover operator. $I(x_{\alpha_1}, x_{\alpha_2}, \ldots)$ and $|\alpha|$ denote the joint mutual information value and the number of selected gene features that are present in both parents from the same module within the NSGA-II mating pool. $I(x_{\beta_1}, x_{\beta_2}, \ldots)$ and $|\beta|$ refer to the joint mutual information value and the number of gene features that were not selected. $I(x_{\gamma_1}, x_{\gamma_2}, \ldots)$ and $|\gamma|$ correspond to the joint mutual information value and the number of gene features that were selected only once from the two parents. It can be observed that when the joint mutual information value of the selected gene features is relatively high and the number of selected features is low, the value of ω will be large, indicating that the selected genes to some extent represent the overall performance of the chosen module. Thus, it can be inferred that the sign of ω is determined by the gene features selected in both parents and those not selected at all. The fewer the selected genes and the higher the joint mutual information value, the greater the likelihood that ω will be positive. The single-selected gene features in the denominator help control the absolute value of ω. When ω is positive, it can be concluded that there are relatively many genes not selected, and their joint mutual information value is small. Therefore, no operations are performed; instead, some single-selected gene features are retained while reducing the number of double-selected gene features to eliminate irrelevant genes that have not been filtered. Conversely, when ω is negative, it is clear that there are too many double-selected genes with a small joint mutual information value. Thus, some double-selected genes are removed, converting them to single-selected or unselected, while adding a few unselected genes to seek a larger joint mutual information value. The value of ω ranges over $\mathscr{R}$ and is designed as a probabilistic operator, implemented through $\omega' = 1 - e^{-|\omega|}$. This design allows the denominator to partially control the magnitude of the metric; a larger value indicates a smaller joint mutual information value for single-selected genes, suggesting that extensive operations are unnecessary. Ultimately, it can be defined that when ω' exceeds a random number between 0 and 1 and ω is positive, transformations are applied to the gene individuals $max(1, \zeta \cdot |\gamma|)$ and $max(1, \zeta \cdot |\alpha|)$ within the module. Conversely, when ω' exceeds a random number between 0 and 1 and ω is negative, transformations are applied to the gene individuals $max(1, \zeta \cdot |\alpha|)$ and $max(1, \zeta \cdot |\beta|)$within the module. The use of the max function is justified by the presence of modules that contain

only a single gene feature. Through the joint mutual information entropy crossover operator, we can identify which genes within M_i are more representative. The crossover process of the CJWGA algorithm is explained in detail in Algorithm 5.

The joint adaptive mutation operator (ϵ) is the second operator proposed in this paper. This operator modifies the population individuals based on multi-criteria evaluation. One type of mutation involves self-modification, which entails transitioning from a selected module M_i to unselected status or from unselected to selected. The second type of mutation focuses on modifying the selection of genes within module M_i. Although the ω operator has performed some evaluative selection from the perspective of the parents, there remains a probability that some relevant genes may be inadvertently overlooked. Therefore, the mutation operator specifically targets the internal gene feature evaluation. Additionally, the mutation process must take into account the number of iterations and the quantity of selected genes within the module. The calculation method for $\epsilon$is as follows:

$$\epsilon = e^{-\left(a^{\delta}-b^{\delta}+log(1+|c|)\right)} \tag{11}$$

In this context, a represents the ratio of the current iteration number to the total iteration number, b denotes the ratio of the number of selected genes in module M_i to the total number of genes, and c is the ratio of the joint mutual information of selected genes to that of the unselected genes. δ is a parameter designed to reduce the influence of the iteration count and the number of selected features on ϵ. This is because the value of gamma should be prioritized, as it potentially reflects the information difference between the selected and unselected features' mutual information in relation to the module. Next, we will discuss how each component affects the specific value of ϵ. As a increases, approaching the final stage of the overall evaluation, the value of ϵwill decrease. At this point, the selection of the module is mostly fixed, and only the genes within module M_i need to be modified. A larger value of b indicates that more genes have been selected within module M_i. In this case, simple self-mutation can be performed, and discarding module M_i can help verify whether better results can be achieved. The impact of c on ϵ is the most significant. It assesses the reliability of the selected genes in module M_i based on the mutual information between selected and unselected gene features. A higher value of c means that the joint mutual information of the unselected genes is larger, indicating a greater need for gene re-selection within M_i with a higher probability. The mutation process of the CJWGA algorithm is explained in detail in Algorithm 6.

By combining information entropy crossover operator and the joint adaptive mutation operator, it is possible to select core modules and perform effective feature extraction within key genes. The pseudocode of the CJWGA algorithm is illustrated in Algorithm 7.

The flowchart of the entire CJWGA algorithm is depicted in Fig. 3, where three distinct modules exert their influence on the outcome of feature selection. The IMGCNet module is located at the top left, the HNSGA-II module with enhanced encoding is at the bottom left, and on the right side are the two proposed crossover and mutation operators.

Computational complexity analysis

To better understand the computational cost of the proposed framework, we analyze the time complexity of each major stage of the algorithm. Let N denote the number of samples, D the number of genes, and T the maximum number of iterations. It is worth noting that in this case, N actually represents the number of modules, so the actual complexity cost is related to the number of selected modules.

In the initial modularization stage, two main computations are required to measure the relationships among genes. First, PCA is used to estimate the structural relationships between modules. The computational complexity of this process is approximately $O\left(N \cdot D^2 + D^3\right)$. Second, mutual information between gene pairs is calculated to evaluate their statistical dependency. Since the mutual information estimation involves pairwise computations across samples, its time complexity is approximately $O\left(N^2 \cdot D\right)$.

After the modules are constructed, the proposed CJWGA algorithm performs feature selection through an evolutionary optimization process. During the population initialization stage, each individual uses a two-layer encoding structure in which both modules and genes within modules are encoded. As a result, the initialization complexity is approximately: $O\left(N^2 \cdot D\right)$ During the evolutionary process, individuals are iteratively updated for T iterations. Considering the population updates across the feature space and module structure, the complexity of the iterative optimization stage can be approximated as: $O\left(N^2 \cdot D \cdot T\right)$.

The proposed operators, namely the CIECO and the JAMO, introduce additional computations during the evolutionary process. For the CIECO operator, the information-based crossover operations involve feature-level calculations within the population, leading to a worst-case time complexity of: $O(N \cdot D \cdot T)$. Similarly, the JAMO operator performs adaptive mutation based on certain threshold conditions. Since the mutation operation is only triggered when the corresponding condition is satisfied, we consider the worst-case scenario where the operator is executed at every iteration. Under this assumption, the complexity of JAMO can also be approximated as: $O(N \cdot D \cdot T)$. However, in practice, the JAMO operator is conditionally triggered and does not execute at every iteration. Therefore, the actual computational overhead introduced by these operators is typically lower than the theoretical worst-case estimation.

Combining the above components, the overall computational complexity of the proposed framework can be summarized as:

$$O\left(N \cdot D^2 + D^3 + N^2 \cdot D \cdot (2+T) + 2 \cdot N \cdot D \cdot T\right)$$

Experimental result

To evaluate the performance of CJWGA in gene data feature selection, we compared it with several other feature selection algorithms and conducted parameter control and ablation experiments. Table 2 provides the basic information of the datasets used in this study, including sample size, labels, and features. It can be observed that all these datasets contain more than 5,000 gene features. The sample sizes vary from 50 to 203, and the number of features ranges from 5,327 to 12,600. The datasets include three categories, with 2 to 4 distinct classes. These variations provide a more comprehensive test of the algorithm's effectiveness and offer a reasonable assessment of its stability and efficiency when handling large-scale data.

To estimate the mutual information and conditional mutual information between genes, we employed an equal-width interval discretization method suitable for small sample sizes. Specifically, the continuous expression values of each gene were discretized into 10 equal-width intervals, dividing the expression range evenly into 10 intervals, and each sample was assigned to the corresponding interval based on its expression value. This parameter achieved a good balance between information retention and statistical stability, conforming to the established practice in gene regulatory

network inference. The discretization method can provide robust estimates of mutual information under small sample conditions, reducing the uncertainty in the estimation.

To evaluate the performance of the feature selection method, we used the K-Nearest Neighbor (KNN) classifier as the classification model. The dataset was randomly divided into a training set and a test set in a $70\% - 30\%$ ratio. To prevent data leakage, all feature selection steps were strictly completed on the training set, and the test set was only used for the final performance evaluation. The final results were reported as the average classification accuracy and standard deviation of the Pareto front solution set obtained from multiple runs, to ensure the stability and reliability of the results.

Fig. 4 presents the clustering dendrograms obtained from WGCNA analysis of six datasets. These are the Brain_Tumor1, Brain_Tumor1, CNS, Leukemia, Leukemia1 and Leukemia2 data sets. The dendrograms allow for a deeper understanding of the gene expression patterns and potential modular structures within the datasets. Datasets such as Brain_Tumor1, Brain_Tumor2, and Leukemia2 exhibit tightly clustered branches, indicating that the genes in these datasets show strong co-expression relationships, suggesting the presence of common biological features or regulatory mechanisms. In contrast, datasets like CNS Leukemia and Leukemia1 show more dispersed clustering, indicating greater variability, which may be due to differences in sample types, processing conditions, or other factors. Regardless of whether the clusters are tight or loose, WGCNA proves to be highly effective. The core objective of WGCNA is to identify gene modules with similar functions or regulatory mechanisms through gene co-expression relationships, and the significance of these modules is not entirely dependent on the tightness of the clustering. Even when certain modules display relatively loose clustering, the genes within those modules may still play critical roles in specific biological processes. On the other hand, tightly clustered modules suggest stronger co-expression relationships between genes, potentially linked to specific biological features or disease states.

All experiments were conducted on a computer with 32 GB of RAM, an i7-12700 CPU@2.10 GHz, using R software version 4.4.1 and MATLAB version 2023a. Furthermore, all experiments in this study were run multiple times, and the final results were computed as the mean and standard deviation. This approach helps mitigate the potential randomness and noise in single-run results, making the conclusions more robust. The population size for all algorithms was set to 50, and the number of evaluations was set to 1,000.

Parameter sensitivity experiment

The performance of an algorithm is often significantly influenced by parameter settings, and different parameter combinations can lead to substantial variations in model performance. By systematically testing and optimizing the parameters, the algorithm's optimal performance in high-dimensional feature selection can be ensured. In this study, two parameters, ζ and δ, are considered. ζ represents the proportion of selected genes within a module in the combined information entropy crossover operator, while δ

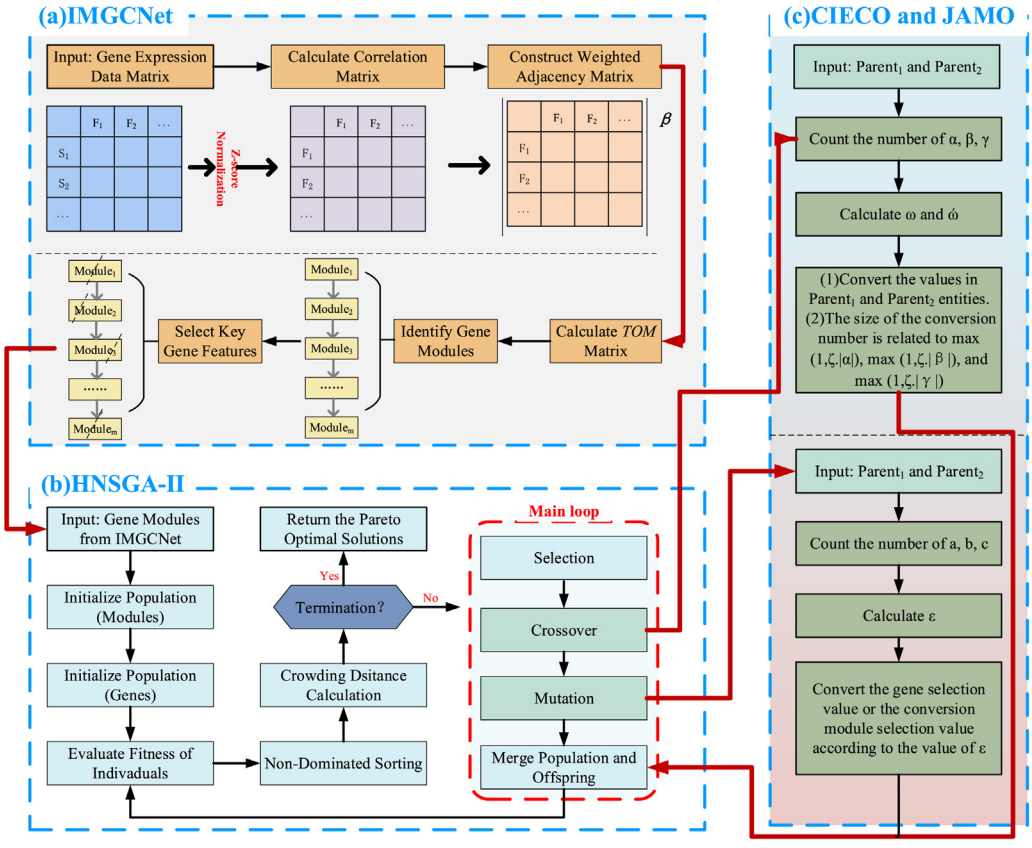


Fig. 3. The flowchart of the entire CJWGA algorithm consists of three modules, each of which has an impact on the results of feature selection. The IMGCNet module for preprocessing is located in the upper left corner, the HNSGA-II module with enhanced encoding function is located in the lower left corner, and on the right side are the two proposed operation operators.

denotes the weight exponent in the joint adaptive mutation operator. In this paper, experiments were conducted with ζ values from the set 0.2, 0.5, 0.8 and δ values from the set 3, 6, 9, using the first six datasets for testing. Tables 3 and 4 show the average classification error rate and the average feature subset ratio for different ζ values. Tables 5 and 6 present the same metrics for different δ values. From Tables 3 and 4, it can be seen that CJWGA1, with ζ set to 0.2, achieves the lowest classification error rate and standard deviation on four datasets, including Brain_Tumor2 and Brain_Tumor1. Regarding feature selection, a ζ value of 0.5 shows greater advantages. When ζ is set to 0.8, the algorithm tends to transform the majority of genes in module M during each iteration, but the experimental results are not competitive.

Overall, a ζ value of 0.2 aligns better with the research goals, as it provides a lower classification error rate, which is particularly crucial for gene-related problems. Tables 5 and 6 present the classification error rates and feature subset ratios for different δ values across the datasets. From Table 5, it can be observed that when δ is set to 3, the exponent has a significant impact on ϵ, resulting in a lower classification error rate. The lowest error rate is achieved on four datasets, with only slight differences on the CNS and Leukemia2 datasets compared to the others. When δ is set to 9, the influence of iteration count and the number of selected genes on ϵ becomes negligible, leading to an increase in classification error rates on certain datasets. Data in Table 6 show that regardless of the δ value, the number of selected genes remains very similar. Considering all factors, a δ value of 3 is the most reasonable, as it ensures a higher precision.

Ablation experiment

To evaluate the overall impact of the combined entropy crossover operator and adaptive mutation operator on the performance of the CJWGA algorithm, two additional GA variants, CWGA and JWGA, were designed. These represent algorithms that use only the combined entropy crossover operator and the adaptive mutation operator, respectively. The tests were conducted on the same datasets, and the results are presented in Tables 7 and 8. In terms of classification error rate, CJWGA achieved the best performance across four datasets, with very small variances, indicating that the algorithm is highly stable. Regarding the proportion of selected feature subsets, CJWGA was able to select the smallest feature subsets on the Leukemia1 and Leukemia2 datasets. In contrast, JWGA, which uses only the mutation operator without the crossover operator, lacks the genetic information from high-quality parents, resulting in minimal changes within the modules. As a result, certain modules were rarely selected throughout the entire iteration, leading to smaller selected subsets. CWGA, on the other hand, performs crossover between the two parents, which emphasizes inter-module variation but is less adept at making detailed gene selections within the module. This experiment demonstrates that both the combined entropy crossover operator and the adaptive mutation operator contribute to the overall convergence of the algorithm. The former focuses more on the global information of the two parents, while the latter emphasizes internal mutations within the modules.

Contrast experiment

In this section, eight publicly available datasets, as described in Table 2, were selected to evaluate the performance of the CJWGA algorithm on discrete combinatorial optimization problems. Three specialized feature selection algorithms, namely FQEISS [63], WMOSS [64], and WQEISS [63], were chosen for comparison. Tables 9 and 10 present the final average results, with the best performance for each dataset highlighted in bold. In terms of classification error rate, CJWGA outperforms all other algorithms on the CNS, Leukemia, Leukemia1, Leukemia2, and Prostate_Tumor datasets. This success is attributed to WGCNA's ability to identify key gene modules early in the process, followed by the two improvement operators, which evaluate the information content of the genes and select key genes that directly correlate with the labels. Table 10 clearly shows that the proposed CJWGA selects the smallest number of genes across all eight datasets. When considering both objectives together, the algorithm ensures optimal performance in either classification error rate or the number of selected genes for all datasets, except for the Lung_Cancer dataset. Moreover, it achieves the best performance in the Leukemia, Leukemia2, and Prostate_Tumor datasets, demonstrating that the best classification accuracy can be achieved with the fewest features.

To further evaluate the multi-objective optimization capability of the proposed method, the distribution of Pareto frontiers obtained by different algorithms was visually presented. Fig. 5 shows the final Pareto frontiers generated by different algorithms after an independent run on all datasets. These Pareto frontiers graphically illustrate the trade-off relationship between the classification error rate and the proportion of selected features. The horizontal axis represents the error rate, and the vertical axis represents the proportion of selected features. As shown in Fig. 5, the Pareto frontiers generated by the proposed CJWGA tend to be superior to those generated by other comparison algorithms. Specifically, many solutions obtained by CJWGA are closer to the lower left corner of the target space in most datasets, indicating that this algorithm can achieve a lower classification error rate while selecting fewer features. In contrast, the Pareto frontiers of several benchmark algorithms are more concentrated, resulting in insufficient diversity, and are farther from the ideal region. This indicates that the trade-off effect achieved between the two objectives is poor.

In addition to this, to quantitatively evaluate the quality of the obtained Pareto fronts, the Inverted Generational Distance (IGD) metric is adopted. In the context of feature selection, the ideal Pareto solution would correspond to a classification error rate of 0 and a feature selection ratio approaching 0. However, such a solution is practically unattainable, since achieving perfect classification with an extremely small number of features is unrealistic for real-world

Table 2
Genetic data set information.

ID	Dataset	Samples	Features	Classes
Dat.1	Brain_Tumor1	90	5920	4
Dat.2	Brain_Tumor2	50	10367	4
Dat.3	CNS	60	7129	2
Dat.4	Leukemia	72	7130	2
Dat.5	Leukemia1	72	5327	3
Dat.6	Leukemia2	70	11225	3
Dat.7	Lung_Cancer	203	12600	3
Dat.8	Prostate_Tumor	102	10509	2

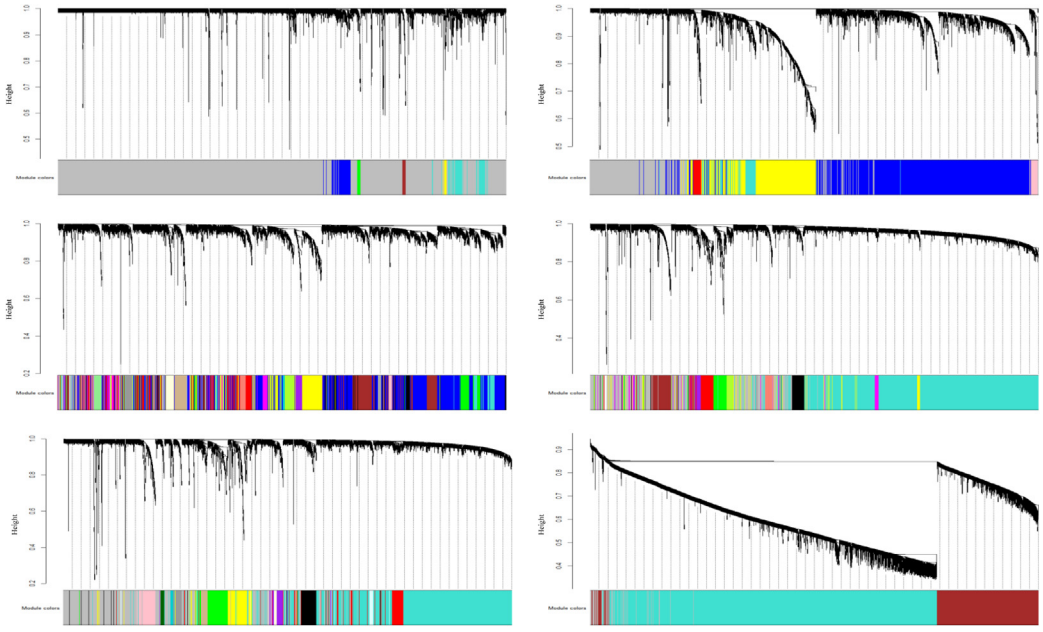


Fig. 4. The hierarchical clustering tree illustrates the hierarchical clustering relationships between samples or features in different datasets. The branching structure of the tree reflects the similarity and grouping patterns at different clustering levels.

Table 3
Statistical table of error rate of parameter sensitivity analysis (1).

Classification error rate	Brain_Tumor1		Brain_Tumor2		CNS	
	mean	std	mean	std	mean	std
CJWGA_ζ1	**2.47852E-01**	**1.33333E-01**	3.49333E-01	1.13333E-01	2.31889E-01	9.44444E-02
CJWGA_ζ2	2.53778E-01	**1.33333E-01**	**2.94267E-01**	1.13333E-01	2.64333E-01	1.05556E-01
CJWGA_ζ3	2.52000E-01	1.59259E-01	3.16933E-01	**1.06667E-01**	**2.16333E-01**	**7.77778E-02**
Classification error rate	Leukemia		Leukemia1		Leukemia2	
	mean	std	mean	std	mean	std
CJWGA_ζ1	**1.43714E-01**	**0.00000E + 00**	**1.60857E-01**	**0.00000E + 00**	**2.02476E-01**	**1.90476E-02**
CJWGA_ζ2	1.67524E-01	**0.00000E + 00**	2.23619E-01	4.76190E-03	2.07048E-01	1.42857E-02
CJWGA_ζ3	1.95714E-01	**0.00000E + 00**	2.06667E-01	**0.00000E + 00**	2.37429E-01	2.38095E-02

Table 4
Parameter sensitivity analysis feature subset proportion statistical Table 1.

Feature subset size	Brain_Tumor1		Brain_Tumor2		CNS	
	mean	std	mean	std	mean	std
CJWGA_ζ1	4.57095E-04	**1.68919E-04**	**2.84943E-04**	**9.64599E-05**	2.63684E-03	8.55660E-04
CJWGA_ζ2	**3.60811E-04**	**1.68919E-04**	3.59410E-04	**9.64599E-05**	2.20760E-03	2.24435E-04
CJWGA_ζ3	3.83446E-04	**1.68919E-04**	4.52783E-04	**9.64599E-05**	**1.04278E-03**	**1.40272E-04**
Feature subset size	Leukemia		Leukemia1		Leukemia2	
	mean	std	mean	std	mean	std
CJWGA_ζ1	1.60982E-03	6.03086E-04	2.28271E-03	5.63169E-04	1.56258E-03	4.09800E-04
CJWGA_ζ2	**8.65077E-04**	1.82328E-04	**1.51455E-03**	3.19129E-04	**8.70557E-04**	1.42539E-04
CJWGA_ζ3	5.68583E-04	**1.40252E-04**	1.60766E-03	**2.06495E-04**	6.73140E-04	**1.15813E-04**

datasets. Therefore, in this study, the reference Pareto front is constructed by combining the non-dominated solutions obtained by all compared algorithms and extracting the global non-dominated set from the union of these solutions. This combined Pareto front is used as the reference set for calculating the IGD values of each algorithm, which provides a fair and widely adopted way to evaluate the convergence and diversity of multi-objective optimization algorithms. Table 11 reports the IGD values obtained by different algorithms on the eight datasets. A smaller IGD value indicates that the obtained Pareto front is closer to the reference Pareto front, which implies better convergence and diversity of the solutions. It can be observed that the proposed CJWGA achieves

Table 5
Statistical table of error rate of parameter sensitivity analysis (2).

Classification error rate	Brain_Tumor1		Brain_Tumor2		CNS	
	mean	std	mean	std	mean	std
CJWGA_δ1	**2.47852E-01**	**1.33333E-01**	**3.43867E-01**	1.06667E-01	2.62556E-01	**7.22222E-02**
CJWGA_δ2	2.53630E-01	1.59259E-01	3.59067E-01	**9.33333E-02**	2.47667E-01	8.88889E-02
CJWGA_δ3	2.49704E-01	1.51852E-01	3.55600E-01	1.00000E-01	**2.36889E-01**	**7.22222E-02**
Classification error rate	Leukemia		Leukemia1		Leukemia2	
	mean	std	mean	std	mean	std
CJWGA_δ1	**1.24190E-01**	**0.00000E + 00**	**2.25155E-03**	**7.50892E-04**	1.98000E-01	**2.38095E-02**
CJWGA_δ2	1.28857E-01	**0.00000E + 00**	1.99048E-01	0.00000E + 00	2.30952E-01	2.85714E-02
CJWGA_δ3	1.59524E-01	**0.00000E + 00**	1.65905E-01	0.00000E + 00	**1.89524E-01**	2.85714E-02

Table 6
Parameter sensitivity analysis feature subset proportion statistical Table 2

Feature subset size	Brain_Tumor1		Brain_Tumor2		CNS	
	mean	std	mean	std	mean	std
CJWGA_δ1	4.57095E-04	**1.68919E-04**	3.15424E-04	**9.64599E-05**	2.23173E-03	5.75116E-04
CJWGA_δ2	**3.78378E-04**	**1.68919E-04**	**2.85328E-04**	**9.64599E-05**	**2.12961E-03**	**4.62898E-04**
CJWGA_δ3	3.97635E-04	**1.68919E-04**	3.12530E-04	**9.64599E-05**	2.62646E-03	8.55660E-04
Feature subset size	Leukemia		Leukemia1		Leukemia2	
	mean	std	mean	std	mean	std
CJWGA_δ1	1.66760E-03	6.31136E-04	2.25155E-03	7.50892E-04	1.80419E-03	6.50334E-04
CJWGA_δ2	1.60589E-03	6.87237E-04	**2.14117E-03**	7.88436E-04	**1.51341E-03**	6.41425E-04
CJWGA_δ3	**1.43534E-03**	**4.90884E-04**	2.26657E-03	**5.81941E-04**	1.62245E-03	**5.16704E-04**

the best performance on most datasets. In particular, CJWGA obtains IGD values of 0 on Brain_Tumor2, CNS, Leukemia, Leukemia1, Leukemia2, and Prostate_Tumor, indicating that its Pareto solutions overlap with the reference Pareto front constructed from all algorithms. This demonstrates that the proposed framework can effectively approximate the optimal trade-off between classification accuracy and feature subset size. Compared with the other algorithms, such as FQEISS, WMOSS, and WQEISS, the proposed CJWGA consistently produces smaller IGD values across most datasets. These results indicate that the proposed method not only achieves better convergence toward the reference Pareto front but also maintains competitive diversity among the obtained solutions, further validating the effectiveness of the proposed multi-objective feature selection framework.

This further highlights the significant impact of the selected genes on the results. The experimental results demonstrate that CJWGA exhibits outstanding optimization performance and exceptional stability, making it a highly effective and reliable feature selection algorithm for medical applications.

In terms of time consumption, the operation methods proposed in this paper, namely CIECO and JAMO, introduce additional computational processes compared to the standard genetic algorithm. However, these computations mainly involve information-theoretic measures within the local gene modules and do not significantly increase the overall computational burden. In fact, the additional costs brought by these operations are much smaller compared to the fitness evaluation based on classifiers, which remains the main factor in the algorithm's running time. Meanwhile, these operations improve the search efficiency by reducing redundant feature combinations and retaining valuable genes, which helps the algorithm converge to a better solution within a reasonable number of iterations.

Regarding the scalability of the method described in this paper, the computational cost of calculating the joint mutual information is the main factor. However, this process is carried out within the gene modules identified in the IMGCNet stage, rather than throughout the entire feature space. Since the size of each module is usually much smaller than the total number of genes, the computational cost of estimating the joint mutual information is significantly reduced. This module-based design enables the proposed framework to maintain reasonable scalability when dealing with high-dimensional microarray data. By limiting the joint informa-

Table 7
Statistical table of ablation experiment error rate.

Classification error rate	Brain_Tumor1		Brain_Tumor2		CNS	
	mean	std	mean	std	mean	std
CJWGA	**2.47852E-01**	**1.33333E-01**	**3.49333E-01**	**1.13333E-01**	**2.31889E-01**	**9.44444E-02**
CWGA	2.58593E-01	1.66667E-01	3.62267E-01	1.46667E-01	2.48000E-01	1.05556E-01
JWGA	2.94296E-01	1.92593E-01	5.23200E-01	2.53333E-01	3.36333E-01	1.66667E-01
Classification error rate	Leukemia		Leukemia1		Leukemia2	
	mean	std	mean	std	mean	std
CJWGA	1.43714E-01	**0.00000E + 00**	**1.60857E-01**	**0.00000E + 00**	2.02476E-01	1.90476E-02
CWGA	**1.23524E-01**	**0.00000E + 00**	1.74095E-01	**0.00000E + 00**	1.73048E-01	**9.52381E-03**
JWGA	2.48857E-01	9.52381E-03	1.62476E-01	**0.00000E + 00**	**1.66762E-01**	1.90476E-02

Table 8
Statistical table of proportion of characteristic subsets in ablation experiments.

Feature subset size	Brain_Tumor1		Brain_Tumor2		CNS	
	mean	std	mean	std	mean	std
CJWGA	4.57095E-04	**1.68919E-04**	2.84943E-04	**9.64599E-05**	2.63684E-03	8.55660E-04
CWGA	4.71622E-04	**1.68919E-04**	3.94135E-04	**9.64599E-05**	3.53766E-03	1.69729E-03
JWGA	**3.60473E-04**	**1.68919E-04**	**2.48288E-04**	**9.64599E-05**	**5.36401E-04**	**1.40272E-04**
Feature subset size	Leukemia		Leukemia1		Leukemia2	
	mean	std	mean	std	mean	std
CJWGA	1.60982E-03	6.03086E-04	**2.28271E-03**	**5.63169E-04**	**1.56258E-03**	**4.09800E-04**
CWGA	2.84208E-03	1.47265E-03	3.26075E-03	1.42669E-03	2.34744E-03	9.97773E-04
JWGA	**7.35764E-04**	**1.40252E-04**	2.65102E-03	9.01070E-04	1.62690E-03	6.85969E-04

Table 9
Error rate results of different algorithms.

Classification error rate	Brain_Tumor1		Brain_Tumor2		CNS		Leukemia	
	mean	std	mean	std	mean	std	mean	std
CJWGA	2.47852E-01	1.33333E-01	3.49333E-01	1.13333E-01	**2.31889E-01**	9.44444E-02	**1.43714E-01**	**0.00000E + 00**
FQEISS	1.67154E-01	**1.57057E-05**	3.21287E-01	**3.59519E-05**	4.76429E-01	1.29538E-04	3.23820E-01	1.22140E-03
WMOSS	**1.63595E-01**	4.48874E-05	**3.04727E-01**	9.41672E-05	4.53258E-01	6.88485E-04	3.13355E-01	7.02791E-04
WQEISS	1.69875E-01	4.73115E-05	3.23087E-01	4.26040E-05	4.75138E-01	**1.22316E-04**	3.40712E-01	4.86669E-04
Classification error rate	Leukemia1		Leukemia2		Lung_Cancer		Prostate_Tumor	
	mean	std	mean	std	mean	std	mean	std
CJWGA	**1.60857E-01**	**0.00000E + 00**	**2.02476E-01**	1.90476E-02	1.58200E-01	4.50000E-02	**2.38667E-01**	2.66667E-02
FQEISS	2.90220E-01	3.12288E-04	3.02542E-01	**1.45896E-04**	1.40276E-01	1.49669E-05	3.42477E-01	1.24570E-03
WMOSS	2.88959E-01	4.54703E-04	3.00191E-01	6.14829E-04	**1.37816E-01**	3.67052E-05	3.64050E-01	1.15100E-03
WQEISS	2.90714E-01	6.90429E-04	3.23256E-01	2.67718E-04	1.41187E-01	**3.22501E-06**	3.96862E-01	**9.57300E-04**

Table 10
Feature subset size results of different algorithms.

Feature subset size	Brain_Tumor1		Brain_Tumor2		CNS		Leukemia	
	mean	std	mean	std	mean	std	mean	std
CJWGA	**4.57095E-04**	**1.68919E-04**	**2.84943E-04**	**9.64599E-05**	2.63684E-03	**8.55660E-04**	**1.60982E-03**	**6.03086E-04**
FQEISS	2.68976E-03	6.73944E-01	1.52220E-03	5.84206E-01	2.21232E-03	3.09146E-01	2.07741E-03	1.84343E + 00
WMOSS	2.31351E-03	3.92894E + 00	1.30430E-03	2.09367E + 00	**1.79735E-03**	2.20360E + 00	2.07172E-03	1.39274E + 00
WQEISS	2.69297E-03	8.04220E-01	1.48529E-03	3.84690E-01	2.19804E-03	1.44260E + 00	2.16285E-03	2.14762E + 00
Feature subset size	Leukemia1		Leukemia2		Lung_Cancer		Prostate_Tumor	
	mean	std	mean	std	mean	std	mean	std
CJWGA	**2.28271E-03**	**5.63169E-04**	1.56258E-03	**4.09800E-04**	2.70524E-03	**8.17460E-04**	**1.01037E-03**	**1.71282E-04**
FQEISS	2.89322E-03	4.05771E-01	1.42290E-03	6.10606E-01	1.26772E-03	7.68054E-01	1.46650E-03	1.25333E + 00
WMOSS	2.53113E-03	9.27611E-01	**1.27587E-03**	2.22567E + 00	**1.10543E-03**	1.16223E + 00	1.30852E-03	4.19053E + 00
WQEISS	2.88143E-03	1.17198E + 00	1.42356E-03	6.27484E-01	1.26988E-03	6.25875E-01	1.46750E-03	1.51760E + 00

tion calculation to local modules, the algorithm avoids the combinatorial explosion phenomenon that may occur if the joint information is calculated on the entire feature set.

Conclusion

This paper proposes the CJWGA feature selection algorithm based on biological networks and information theory. It also improves the crossover and mutation operators in NSGA-II by introducing the combined entropy crossover operator and adaptive mutation operator to address the high-dimensional feature selection problem in gene data. By calculating the joint mutual information values for the selection of genes within different modules, the algorithm assesses the information content of genes and more accurately selects key features. The experimental results demonstrate that the CJWGA algorithm can achieve precise classification tasks with a small number of effective genes across multiple high-dimensional and complex medical gene datasets. Although the proposed framework has demonstrated excellent performance in identifying compact and predictive gene subsets, in this study, the biological explanations for the selected genes have not been fully explored. Therefore, a systematic biological interpretation of the selected genes will be regarded as an important direction for future work. In addition, the proposed CJWGA algorithm can not only be applied to medical gene feature selection, but also to dimensionality reduction and other problems in various fields.

Declaration of generative AI

During the preparation of this work the author(s) used Doubao Pro ([https://www.doubao.com/]) in order to polish the language and logical flow of the manuscript. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

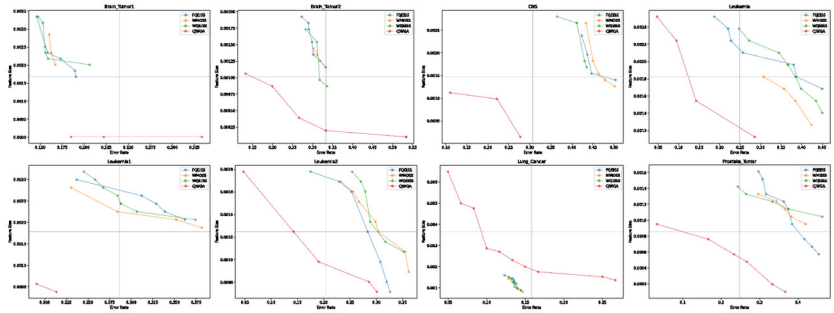


Figure 5: Distribution of pareto front: Each subfigure corresponds to one dataset. The horizontal axis represents the classification error rate, while the vertical axis indicates the proportion of selected features. The Pareto fronts illustrate the trade-off between minimizing classification error and reducing the number of selected features.

Table 11
The IGD values of different algorithms on the dataset

	FQEISS	WMOSS	WQEISS	CJWGA
Brain_Tumor1	0.0219	0.0301	0.0194	**0.0193**
Brain_Tumor2	0.1035	0.1248	0.1077	**0.0000**
CNS	0.2196	0.2296	0.1613	**0.0000**
Leukemia	0.0764	0.1641	0.1174	**0.0000**
Leukemia1	0.0348	0.0283	0.0431	**0.0000**
Leukemia2	0.0368	0.0663	0.0791	**0.0000**
Lung_Cancer	0.0167	0.0197	0.0181	**0.0027**
Prostate_Tumor	0.0848	0.0828	0.0512	**0.0000**

proposed CJWGA achieves the best performance on most datasets. In particular, CJWGA obtains IGD values of 0 on Brain_Tumor2, CNS, Leukemia, Leukemia1, Leukemia2, and Prostate_Tumor, indicating that its Pareto solutions overlap with the reference Pareto front constructed from all algorithms. This demonstrates that the proposed framework can effectively approximate the optimal trade-off between classification accuracy and feature subset size. Compared with the other algorithms, such as FQEISS, WMOSS, and WQEISS, the proposed CJWGA consistently produces smaller IGD values across most datasets. These results indicate that the proposed method not only achieves better convergence toward the reference Pareto front but also maintains competitive diversity among the obtained solutions, further validating the effectiveness of the proposed multi-objective feature selection framework.

This further highlights the significant impact of the selected genes on the results. The experimental results demonstrate that CJWGA exhibits outstanding optimization performance and exceptional stability, making it a highly effective and reliable feature selection algorithm for medical applications.

In terms of time consumption, the operation methods proposed in this paper, namely CIECO and JAMO, introduce additional computational processes compared to the standard genetic algorithm. However, these computations mainly involve information-theoretic measures within the local gene modules and do not significantly increase the overall computational burden. In fact, the additional costs brought by these operations are much smaller compared to the fitness evaluation based on classifiers, which remains the main factor in the algorithm's running time. Meanwhile, these operations improve the search efficiency by reducing redundant feature combinations and retaining valuable genes, which helps the algorithm converge to a better solution within a reasonable number of iterations.

Regarding the scalability of the method described in this paper, the computational cost of calculating the joint mutual information is the main factor. However, this process is carried out within the gene modules identified in the IMGCNet stage, rather than throughout the entire feature space. Since the size of each module is usually much smaller than the total number of genes, the computational cost of estimating the joint mutual information is significantly reduced. This module-based design enables the proposed framework to maintain reasonable scalability when dealing with high-dimensional microarray data. By limiting the joint information calculation to local modules, the algorithm avoids the combinatorial explosion phenomenon that may occur if the joint information is calculated on the entire feature set.

6. Conclusion

This paper proposes the CJWGA feature selection algorithm based on biological networks and information theory. It also improves the crossover and mutation operators in NSGA-II by introducing the combined entropy crossover operator and adaptive mutation operator to address the high-dimensional feature selection problem in gene data. By calculating the joint mutual information values for the selection of genes within different modules, the algorithm assesses the information content of genes and more accurately selects key features. The experimental results demonstrate that the CJWGA algorithm can achieve precise classification tasks with a small number of effective genes across multiple high-dimensional and complex medical gene datasets. Although

Fig. 5. Distribution of Pareto front: Each subfigure corresponds to one dataset. The horizontal axis represents the classification error rate, while the vertical axis indicates the proportion of selected features. The Pareto fronts illustrate the trade-off between minimizing classification error and reducing the number of selected features.

Table 11
The IGD values of different algorithms on the dataset.

	FQEISS	WMOSS	WQEISS	CJWGA
Brain_Tumor1	0.0219	0.0301	0.0194	**0.0193**
Brain_Tumor2	0.1035	0.1248	0.1077	**0.0000**
CNS	0.2196	0.2296	0.1613	**0.0000**
Leukemia	0.0764	0.1641	0.1174	**0.0000**
Leukemia1	0.0348	0.0283	0.0431	**0.0000**
Leukemia2	0.0368	0.0663	0.0791	**0.0000**
Lung_Cancer	0.0167	0.0197	0.0181	**0.0027**
Prostate_Tumor	0.0848	0.0828	0.0512	**0.0000**

Compliance with ethics requirements

This article does not contain any studies with human or animal subjects.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

[1] Stevenson Jonathan. Self-destruct sequence. Commonweal 2011;138(9):36.
[2] Cañas Benito, López-Ferrer Daniel, Ramos-Fernández Antonio, Camafeita Emilio, Calvo Enrique. Mass spectrometry technologies for proteomics. Brief Funct Genom 2006;4(4):295–320.
[3] Molster Caron M, Bowman Faye L, Bilkey Gemma A, Cho Angela S, Burns Belinda L, Nowak Kristen J, Dawkins Hugh JS. The evolution of public health genomics: exploring its past, present, and future. Front Public Health 2018;6:247.
[4] Alexander Ebhardt H, Root Alex, Sander Chris, Aebersold Ruedi. Applications of targeted proteomics in systems biology and translational medicine. Proteomics 2015;15(18):3193–208.
[5] Yan Min, Guowang Xu. Current and future perspectives of functional metabolomics in disease studies–a review. Anal Chim Acta 2018;1037:41–54.
[6] Dhanuka Richa, Singh Jyoti Prakash, Tripathi Anushree. A comprehensive survey of deep learning techniques in protein function prediction. IEEE/ACM Trans Comput Biol Bioinf 2023;20(3):2291–301.
[7] Bamunu Mudiyanselage Thosini K, Xiao Xueli, Zhang Yanqing, Pan Yi. Deep fuzzy neural networks for biomarker selection for accurate cancer detection. IEEE Trans Fuzzy Syst 2019;28(12):3219–28.
[8] Daniel L Cook, Anthony N Gerber, and Stephen J Tapscott. Modeling stochastic gene expression: implications for haploinsufficiency. Proceedings of the National Academy of Sciences, 95(26), 15641–15646, 1998.
[9] Bunker Grant, Dimakis Nicholas, Khelashvili Gocha. New methods for exafs analysis in structural genomics. Synchrotr Radiat 2005;12(1):53–6.
[10] Sucre Aurora, Martinez Maria, Garin-Muga Alba. Omicsdk-transcriptomics: a web platform for transcriptomics data analysis. In: Caring is sharing-exploiting the value in data for health and innovation. IOS Press; 2023. p. 1042–6.
[11] Guilherme Reis-de Oliveira, Victor Corasolla Carregari, Gabriel Rodrigues dos Reis de Sousa, and Daniel Martins-de Souza. Omicscope unravels systems-level insights from quantitative proteomics data. Nature Communications, 15 (1):6510, 2024.
[12] Seyednami Niyakan, Jianting Sheng, Yuliang Cao, Xiang Zhang, Zhan Xu, Ling Wu, Stephen TC Wong, and Xiaoning Qian. Mustang: Multi-sample spatial transcriptomics data analysis with cross-sample transcriptional similarity guidance. Patterns, 5(5), 2024.
[13] Zhang Jingyi, Ma Ping, Zhong Wenxuan, Meng Cheng. Projection-based techniques for high-dimensional optimal transport problems. Wiley Interdiscip Rev: Comput Stat 2023;15(2):e1587.
[14] Yang Geying, He Junjiang, Lan Xiaolong, Li Tao, Fang Wenbo. A fast dual-module hybrid high-dimensional feature selection algorithm. Inf Sci 2024;681:121185.
[15] Wang Xubin, Shangguan Haojiong, Huang Fengyi, Shangrui Wu, Jia Weijia. Mel: efficient multi-task evolutionary learning for high-dimensional feature selection. IEEE Trans Knowl Data Eng 2024.
[16] Unler Alper, Murat Alper. A discrete particle swarm optimization method for feature selection in binary classification problems. Eur J Oper Res 2010;206 (3):528–39.
[17] Liu Yuanning, Wang Gang, Chen Huiling, Dong Hao, Zhu Xiaodong, Wang Sujing. An improved particle swarm optimization for feature selection. J Bionic Eng 2011;8(2):191–200.
[18] Zhang Lei, Sun Xiang, Yang Haipeng, Cheng Fan. Sparsity preserved pareto optimization for subset selection. IEEE Trans Evol Comput 2023.
[19] Gao Yunlong, Qinting Wu, Zhenghong Xu, Cao Chao, Pan Jinyan, Shao Guifang, Nie Feiping, Zhu Qingyuan. Principal component analysis with fuzzy elastic net for feature selection. IEEE Trans Fuzzy Syst 2024;32(12):6878–90.
[20] Zaman Ezzatul Akmal Kamaru, Mohamed Azlinah, Ahmad Azlin. Feature selection for online streaming high-dimensional data: a state-of-the-art review. Appl Soft Comput 2022;127:109355.
[21] Matheus Cezimbra Barbieri, Bruno Iochins Grisci, and Márcio Dorn. Analysis and comparison of feature selection methods towards performance and stability. Expert Systems with Applications, page 123667, 2024.
[22] Zhang Hengyi. Feature selection for single cell rna sequencing data based on a noise-robust fuzzy relation and fuzzy evidence theory. Appl Soft Comput 2023;148:110940.
[23] Xiaoqin Ma, Jianming Liu, Pei Wang, Wenchang Yu, and Huanhuan Hu. Feature selection for hybrid information systems based on fuzzy β covering and fuzzy evidence theory. Journal of Intelligent & Fuzzy Systems, (Preprint):1–24, 2024.
[24] Huang HaiHui, NaiQi Wu, Liang Yong, Peng XinDong, Shu Jun. Slnl: a novel method for gene selection and phenotype classification. Int J Intell Syst 2022;37(9):6283–304.
[25] Tsai Chih-Fong, Sung Ya-Ting. Ensemble feature selection in high dimension, low sample size datasets: parallel and serial combination approaches. Knowl -Based Syst 2020;203:106097.
[26] Dash Manoranjan, Liu Huan. Feature selection for classification. Intell Data Anal 1997;1(1–4):131–56.
[27] Guyon Isabelle, Elisseeff André. An introduction to variable and feature selection. J Mach Learn Res 2003;3(Mar):1157–82.
[28] Albukhanajer Wissam A, Jin Yaochu, Briffa Johann A, Williams Godfried. Evolutionary multi-objective optimization of trace transform for invariant feature extraction. In: 2012 IEEE congress on evolutionary computation. IEEE; 2012. p. 1–8.
[29] He Binsheng, Lang Jidong, Wang Bo, Liu Xiaojun, Qingqing Lu, He Jianjun, Gao Wei, Bing Pingping, Tian Geng, Yang Jialiang. Toome: a novel computational framework to infer cancer tissue-of-origin by integrating both gene mutation and expression. Front Bioeng Biotechnol 2020;8:394.
[30] Lazar Cosmin, Taminau Jonatan, Meganck Stijn, Steenhoff David, Coletta Alain, Molter Colin, de Schaetzen Virginie, Duque Robin, Bersini Hugues, Nowe Ann. A survey on filter techniques for feature selection in gene expression microarray analysis. IEEE/ACM Trans Comput Biol Bioinform 2012;9 (4):1106–19.
[31] Kwak Nojun, Choi Chong-Ho. Input feature selection for classification problems. IEEE Trans Neural Networks 2002;13(1):143–59.
[32] Unler Alper, Murat Alper, Chinnam Ratna Babu. mr2pso: a maximum relevance minimum redundancy feature selection method based on swarm intelligence for support vector machine classification. Inf Sci 2011;181 (20):4625–41.
[33] Uda Shinsuke, Kuroda Shinya. Analysis of cellular signal transduction from an information theoretic approach. Seminars in cell & developmental biology, volume 51. Elsevier; 2016. p. 24–31.

[34] Tibshirani Robert. Regression shrinkage and selection via the lasso. J Roy Stat Soc Ser B: Stat Methodol 1996;58(1):267–88.
[35] Chen Tianqi, Guestrin Carlos. Xgboost: a scalable tree boosting system. In: Proceedings of the 22nd acm sigkdd international conference on knowledge discovery and data mining. p. 785–94.
[36] Zhang Bin, Horvath Steve. A general framework for weighted gene co-expression network analysis. Stat Appl Genet Mol Biol 2005;4(1).
[37] Stuart Joshua M, Segal Eran, Koller Daphne, Kim Stuart K. A gene-coexpression network for global discovery of conserved genetic modules. Science 2003;302 (5643):249–55.
[38] MacLennan Nicole K, Dong Jun, Aten Jason E, Horvath Steve, Rahib Lola, Ornelas Loren, Dipple Katrina M, McCabe Edward RB. Weighted gene co-expression network analysis identifies biomarkers in glycerol kinase deficient mice. Mol Genet Metabol 2009;98(1–2):203–14.
[39] Wang Yadi, Li Xiaoping, Ruiz Rubén. Weighted general group lasso for gene selection in cancer classification. IEEE Trans Cybernet 2018;49 (8):2860–73.
[40] Xie Ming, Li Xiandeng, Qi Congwei, Zhang Yufeng, Li Gang, Xue Yong, Chen Guobao. Feature genes identification and immune infiltration assessment in abdominal aortic aneurysm using wgcna and machine learning algorithms. Front Cardiovasc Med 2024;11:1497170.
[41] Liu Huan, Zhao Zheng. Manipulating data and dimension reduction methods: Feature selection. Computational complexity: theory, techniques, and applications. New York: Springer; 2012. p. 1790–800.
[42] Huan Liu, Hiroshi Motoda, Rudy Setiono, and Zheng Zhao. Feature selection: An ever evolving frontier in data mining. In Feature selection in data mining, pages 4–13. PMLR, 2010.
[43] Solorio-Fernandez Saul, Martínez-Trinidad José Fco, Ariel Carrasco-Ochoa J. A supervised filter feature selection method for mixed data based on spectral feature selection and information-theory redundancy analysis. Pattern Recogn Lett 2020;138:321–8.
[44] Bommert Andrea, Welchowski Thomas, Schmid Matthias, Rahnenführer Jörg. Benchmark of filter methods for feature selection in high-dimensional gene expression survival data. Brief Bioinform 2022;23(1):bbab354.
[45] George Obaido, Okechinyere Achilonu, Blessing Ogbuokiri, Chimeremma Sandra Amadi, Lawal Habeebullahi, Tony Ohalloran, Chidozie Williams Chukwu, Ebikella Mienye, Mikail Aliyu, Olufunke Fasawe, et al. An improved framework for detecting thyroid disease using filter-based feature selection and stacking ensemble. IEEE Access, 2024.
[46] Xue Bing, Cervante Liam, Shang Lin, Browne Will N, Zhang Mengjie. Multi-objective evolutionary algorithms for filter based feature selection in classification. Int J Artif Intell Tools 2013;22(04):1350024.
[47] Medjahed Seyyid Ahmed, Saadi Tamazouzt Ait, Benyettou Abdelkader, Ouali Mohammed. Kernel-based learning and feature selection analysis for cancer diagnosis. Appl Soft Comput 2017;51:39–48.
[48] Luo Chuan, Wang Sizhao, Li Tianrui, Chen Hongmei, Lv Jiancheng, Yi Zhang. Large-scale meta-heuristic feature selection based on bpso assisted rough hypercuboid approach. IEEE Trans Neural Networks Learn Syst 2022;34 (12):10889–903.
[49] Hashemi Amin, Joodaki Mehdi, Joodaki Nazanin Zahra, Dowlatshahi Mohammad Bagher. Ant colony optimization equipped with an ensemble of heuristics through multi-criteria decision making: a case study in ensemble feature selection. Appl Soft Comput 2022;124:109046.
[50] Ibrahim Rehab Ali, Ewees Ahmed A, Oliva Diego. Mohamed Abd Elaziz, and Songfeng Lu. Improved salp swarm algorithm based on particle swarm optimization for feature selection. J Ambient Intell Human Comput 2019;10:3155–69.
[51] H Hannah Inbarani, Ahmad Taher Azar, and G Jothi. Supervised hybrid feature selection based on pso and rough sets for medical diagnosis. Computer methods and programs in biomedicine, 113(1):175–185, 2014.
[52] Chen Dingiun, Chan Keith CC, Xindong Wu. Gene expression analyses using genetic algorithm based hybrid approaches. In: 2008 IEEE Congress on Evolutionary Computation (IEEE World Congress on Computational Intelligence). IEEE; 2008. p. 963–9.
[53] Joaquín Derrac, Salvador García, and Francisco Herrera. A first study on the use of coevolutionary algorithms for instance and feature selection. In Hybrid Artificial Intelligence Systems: 4th International Conference, HAIS 2009, Salamanca, Spain, June 10–12, 2009. Proceedings 4, pages 557–564. Springer, 2009.
[54] Mukhopadhyay Anirban, Maulik Ujjwal. An svm-wrapped multiobjective evolutionary feature selection approach for identifying cancer-microrna markers. IEEE Trans Nanobioscience 2013;12(4):275–81.
[55] Jović Alan, Brkić Karla, Bogunović Nikola. A review of feature selection methods with applications. In: 2015 38th international convention on information and communication technology, electronics and microelectronics (MIPRO). Ieee; 2015. p. 1200–5.
[56] Nematzadeh Hossein, García-Nieto José, Aldana-Montes José F, Navas-Delgado Ismael. Pattern recognition frequency-based feature selection with multi-objective discrete evolution strategy for high-dimensional medical datasets. Exp Syst Appl 2024;249:123521.
[57] Zhaozhao Xu, Yang Fangyuan, Tang Chaosheng, Wang Hong, Wang Shuihua, Sun Junding, Zhang Yudong. Fg-hfs: a feature filter and group evolution hybrid feature selection algorithm for high-dimensional gene expression data. Exp Syst Appl 2024;245:123069.
[58] Ashis Kumar Mandal, Md Nadim, Hasi Saha, Tangina Sultana, Md Delowar Hossain, and Eui-Nam Huh. Feature subset selection for high-dimensional, low sampling size data classification using ensemble feature selection with a wrapper-based search. IEEE Access, 2024.
[59] Bir-Jmel Ahmed, Douiri Sidi Mohamed, Elbernoussi Souad. Minimum redundancy maximum relevance and vns based gene selection for cancer classification in high-dimensional data. Int J Comput Sci Eng 2024;27 (1):78–89.
[60] Wong William WL, Burkowski Forbes J. Using kernel alignment to select features of molecular descriptors in a qsar study. IEEE/ACM Trans Comput Biol Bioinf 2011;8(5):1373–84.
[61] Wang Yingying, Fan Xiaomao, Cai Yunpeng. A comparative study of improvements pre-filter methods bring on feature selection using microarray data. Health Inform Sci Syst 2014;2:1–9.
[62] Deb Kalyanmoy et al. A fast and elitist multiobjective genetic algorithm: NSGA-II. IEEE Trans Evol Comput 2002;6(2):182–97.
[63] Karakaya Gülşah, Galelli Stefano, Ahipaşaoğlu Selin Damla, Taormina Riccardo. Identifying (quasi) equally informative subsets in feature selection problems for classification: a max-relevance min-redundancy approach. IEEE Trans Cybernet 2015;46(6):1424–37.
[64] Hamdani Tarek M, Won Jin-Myung, Alimi Adel M, Karray Fakhri. Multi-objective feature selection with nsga ii. In: Adaptive and natural computing algorithms: 8th international conference, ICANNGA 2007, Warsaw, Poland, April 11–14; 2007, Proceedings, Part I 8, pages 240–247. Springer, 2007.